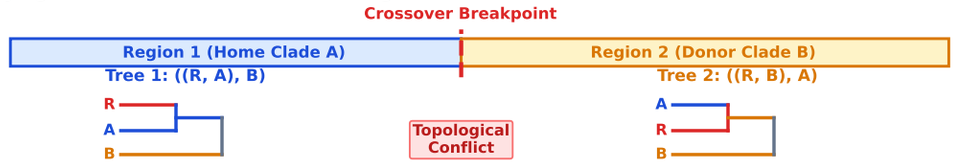
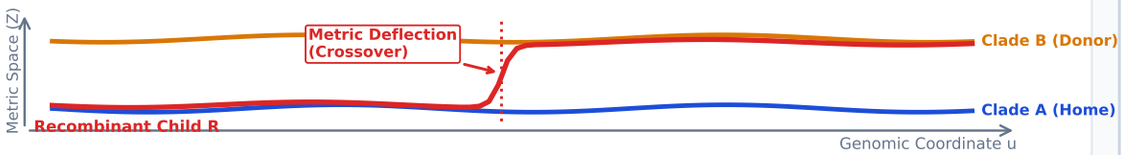
A From Bifurcating Trees to Continuous Trajectories

Recombination shatters trees into discordant segments; trajectories model reticulation as metric flow.

Phylogenetic Discordance Shatters Single-Tree Models



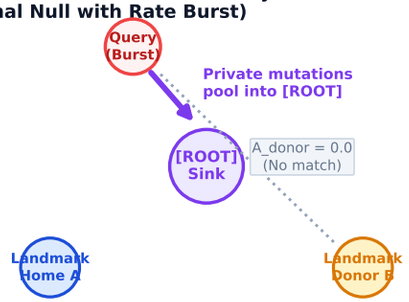
Continuous Trajectory Mechanics in Metric Space



B Phylogenetic Cross-Attention & Autapomorphy Root Sink

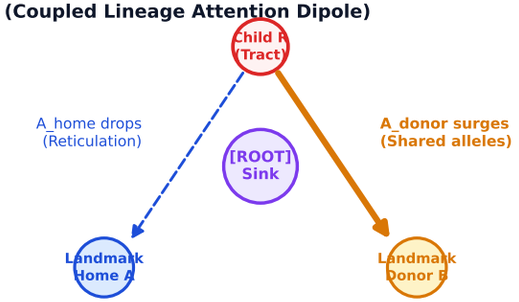
The unpolarized [ROOT] sink absorbs lineage rate bursts, preventing heuristic false alarms.

Mutational Burst / Heterotachy (Clonal Null with Rate Burst)



False Positive Rate = 0.0%
Rate burst absorbed by sink

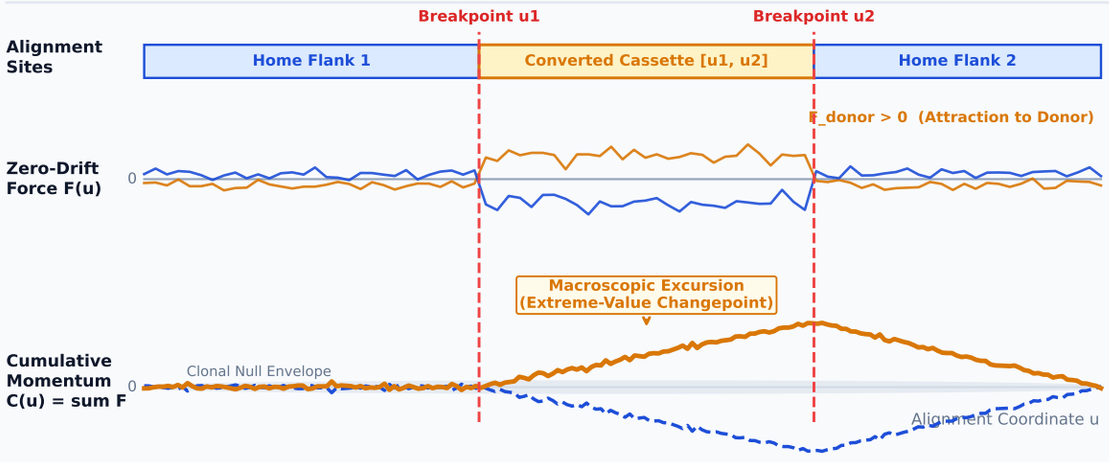
Authentic Recombination Event (Coupled Lineage Attention Dipole)



Anti-Correlated Dipole
Delta A_donor > 0 & Delta A_home < 0

C Physical Signal Deconvolution & Momentum Trajectories

Zero-drift force integration converts noisy site attention into macroscopic Brownian excursions.



D Resolution Across Evolutionary Regimes & Swarm Consolidation

Deconvolving epidemic transmission swarms, ghost donors, and micro-conversions at genome scale.

Epidemic Transmission Swarm Consolidation

Triplet Heuristics (3Seq / RDP5):

Evaluates each isolate independently

50 Isolates = 50 Flagged Events!
(10x Overcounting Inflation)

→

RhizAeon Graph Adjudication:

Retains shared ancestral history

1 Unified Ancestral Event
(50 Descendant Isolates linked)

Deconvolution Across Empirical Genomic Architectures

Viral Chimeras (Sarbecovirus)	Bacterial Chromosomes (S. pneumoniae)	Immunogenomics (Primate HLA)
Multi-donor mosaics Ghost donors SARS-CoV-2 Spike	SNP coordinates PMEN1 penicillin pbp2x 96.2% Gubbins match	Inter-locus conversions Peptide binding groove HLA-B*46:01 chimera
30 kb / 83 taxa	2.22 Mb in <10 ms	50 bp micro-tracts